

Recall from last class:

- For a matrix  $A$  define  $A_{ij}$  to be the matrix obtained from  $A$  by deleting row  $i$  and column  $j$ .
- For  $n \geq 2$ , the determinant of an  $n \times n$  matrix  $A = [a_{ij}]$  is

$$\det(A) = a_{11} \det(A_{11}) - a_{12} \det(A_{12}) + \cdots + (-1)^{n+1} a_{1n} \det(A_{1n}) = \sum_{j=1}^n (-1)^{1+j} a_{1j} \det(A_{1j}).$$

We call this the cofactor expansion across the first row.

- **Mohan's Theorem:** Cofactor expansion across any row or column is the same, and hence each cofactor expansion gives the determinant.

1. Compute the determinant of  $C = \begin{bmatrix} 6 & 0 & 0 & 5 \\ 1 & 7 & 2 & -5 \\ 2 & 0 & 0 & 0 \\ 8 & 3 & 1 & 8 \end{bmatrix}$ .

**Definition 1.** Let  $A = [a_{ij}]$  be an  $n \times n$  matrix, then  $A$  is **upper triangular** if  $a_{ij} = 0$  when  $i > j$ .

**Theorem 3.2:** Let  $A$  be an  $n \times n$  upper triangular matrix, then

$$\det(A) = a_{11}a_{22} \dots a_{nn} = \prod_{j=1}^n a_{jj}.$$

2. Prove the theorem.

**Goal:** Prove  $\det(A) \neq 0$  if and only if  $A$  is invertible.

**Theorem 3.3:** If  $A$  is an  $n \times n$  matrix and  $E$  is an elementary matrix, then

$$\det(EA) = \det(E) \det(A),$$

and

$$\det(E) = \begin{cases} 1 & \text{if } E \text{ represents adding a multiple of a row to another row,} \\ -1 & \text{if } E \text{ represents interchanging two rows,} \\ r & \text{if } E \text{ represents scaling a row by a factor of } r. \end{cases}$$

**3.** Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad E_1 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad E_3 = \begin{bmatrix} 10 & 0 \\ 0 & 1 \end{bmatrix}.$$

Check the theorem for these matrices.

4. Prove the theorem.

**Theorem 3.6:** If  $A$  and  $B$  are  $n$  by  $n$  matrices, then

$$\det(AB) = \det(A) \det(B).$$

5. Prove the theorem.

6. **Corollary:**  $A$  is invertible if and only if  $\det(A) \neq 0$ . Prove the corollary.

**7. True or False:**

(a) If  $A$  is an  $n \times n$  invertible matrix, then  $\det(A^{-1}) = \frac{1}{\det(A)}$ .

(b) If  $A$  is an  $n \times n$  matrix, then  $\det(A) = \det(A^T)$ .

(c) If  $A \sim B$ , then  $\det(A) = \det(B)$ .

**8.** We can think of  $\det$  as a function from the set of  $n \times n$  matrices to the real numbers. Is this function linear?

# Determinant Properties – Answers / Solutions

1. Again we use the definition to compute the determinant:

$$\det C = \det \begin{bmatrix} 6 & 0 & 0 & 5 \\ 1 & 7 & 2 & -5 \\ 2 & 0 & 0 & 0 \\ 8 & 3 & 1 & 8 \end{bmatrix} = 6 \cdot \det \begin{bmatrix} 7 & 2 & -5 \\ 0 & 0 & 0 \\ 3 & 1 & 8 \end{bmatrix} - 5 \cdot \det \begin{bmatrix} 1 & 7 & 2 \\ 2 & 0 & 0 \\ 8 & 3 & 1 \end{bmatrix} = -5(-2) = 10.$$

Using the last Theorem, we can expand about the third row as well. We show that here:

$$\det C = \det \begin{bmatrix} 6 & 0 & 0 & 5 \\ 1 & 7 & 2 & -5 \\ 2 & 0 & 0 & 0 \\ 8 & 3 & 1 & 8 \end{bmatrix} = 2 \cdot \det \begin{bmatrix} 0 & 0 & 5 \\ 7 & 2 & -5 \\ 3 & 1 & 8 \end{bmatrix} = 2(5)(7 - 6) = 10.$$

2. *Proof.* We prove the result by induction on the size of the matrix.

**Base Case:** Let  $A = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$ , then  $\det(A) = ac$ . Thus the result holds for  $2 \times 2$  matrices.

**Inductive Step:** We assume that the determinant of an  $n \times n$  upper triangular matrix is given by the product of the diagonal elements.

Let  $A$  be an  $(n + 1) \times (n + 1)$  upper triangular matrix. We compute the determinant by expanding along the first column:

$$\det(A) = a_{11} \det(A_{11}) - a_{21} \det(A_{21}) + \cdots + (-1)^{n+1+1} a_{(n+1)1} \det(A_{(n+1)1})$$

Since  $A$  is upper-triangular,  $a_{21} = \cdots = a_{(n+1)1} = 0$ . Thus

$$\det(A) = a_{11} \det(A_{11}).$$

Since  $A_{11}$  is an  $n \times n$  upper triangular matrix, we can apply the induction hypothesis to get

$$\det(A) = a_{11} \det(A_{11}) = a_{11}(a_{22} \cdots a_{(n+1)(n+1)}).$$

Thus the result holds by mathematical induction. □

3. We first compute the products:

$$\begin{aligned} E_1 A &= \begin{bmatrix} 1 & 2 \\ 5 & 8 \end{bmatrix}, \\ E_2 A &= \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}, \\ E_3 A &= \begin{bmatrix} 10 & 20 \\ 3 & 4 \end{bmatrix}. \end{aligned}$$

Thus notice

$$\begin{aligned} \det E_1 A &= 8 - 10 = -2 = \det E_1 \det A, \\ \det E_2 A &= 6 - 4 = 2 = \det E_2 \det A, \\ \det E_3 A &= 40 - 60 = -20 = \det E_3 \det A. \end{aligned}$$

Thus the Theorem seems to work in all these cases.

4. We prove the result by induction on the size of the matrix  $A$ .

**Base Cases:** Let  $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ .

**Case 1:** Suppose  $E$  corresponds to a 2 by 2 matrix scaling a row by a factor of  $r$ , then  $E = \begin{bmatrix} r & 0 \\ 0 & 1 \end{bmatrix}$  or  $E = \begin{bmatrix} 1 & 0 \\ 0 & r \end{bmatrix}$ . Both matrices are similar, so we only check the first.

Note  $\det EA = \det \begin{bmatrix} ra & rb \\ c & d \end{bmatrix} = rad - rbc = r(ad - bc) = \det E \det A$ , and hence the result holds in this case.

**Case 2:** Suppose  $E$  corresponds to a 2 by 2 matrix interchanging rows, then  $E = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ .

Note  $\det EA = \det \begin{bmatrix} c & d \\ a & b \end{bmatrix} = bc - ad = -(ad - bc) = \det E \det A$ , and hence the result holds in this case.

**Case 3:** Suppose  $E$  corresponds to a 2 by 2 matrix adding  $r$  times a row to another row, then  $E = \begin{bmatrix} 1 & r \\ 0 & 1 \end{bmatrix}$  or  $E = \begin{bmatrix} 1 & 0 \\ r & 1 \end{bmatrix}$ . Both matrices are similar, so we only check the first.

Note  $\det EA = \det \begin{bmatrix} a + rc & b + rd \\ c & d \end{bmatrix} = (a + rc)d - c(b + rd) = ad - bc = \det E \det A$ , and hence the result holds in this case.

**Inductive Step:** Suppose the result holds for  $n$  by  $n$  matrices. Let  $A$  be an  $(n + 1) \times (n + 1)$  matrix and let  $E$  be an elementary matrix. Since  $n \geq 2$  and elementary row operations change at most 2 rows, we know there must be some row of  $EA$  that is unchanged. We call that row  $k$  and expand the determinant along that row. Let  $C = EA$ , then  $C_{kj}$  is an  $n$  by  $n$  submatrix related to  $A_{kj}$  by the same elementary row operation as  $E$ . By the induction hypothesis, we get  $\det C_{kj} = \det E \det A_{kj}$  for all  $1 \leq j \leq n + 1$ . Now we compute the determinant of  $EA$  expanding along row  $k$ , then

$$\begin{aligned} \det EA &= \det C = (-1)^{k+1} a_{k1} \det C_{k1} + \cdots + (-1)^{n+1+k} a_{k(n+1)} \det C_{k(n+1)} \\ &= (-1)^{k+1} a_{k1} \det E \det A_{k1} + \cdots + (-1)^{n+1+k} a_{k(n+1)} \det E \det A_{k(n+1)} \\ &= \det E \left( (-1)^{k+1} a_{k1} \det A_{k1} + \cdots + (-1)^{n+1+k} a_{k(n+1)} \det A_{k(n+1)} \right) \\ &= \det E \det A. \end{aligned}$$

Thus by mathematical induction the result holds for all  $n$  by  $n$  matrices.

5. *Proof.* We prove the result by cases.

**Case 1:**  $A$  is invertible.

Since  $A$  is invertible, the invertible matrix theorem tells us that there exists elementary row operations  $E_1, \dots, E_p$  such that  $I_n = E_p \dots E_1 A$ . Since elementary matrices are invertible and the inverse of an elementary matrix is another elementary matrix, we get  $A = E_1^{-1} \dots E_p^{-1}$ .



Thus we get  $AB = E_1^{-1} \dots E_p^{-1}B$ . Now we apply the previous theorem multiple times to get the following

$$\begin{aligned}\det AB &= \det E_1^{-1} \dots E_p^{-1}B \\ &= \det E_1^{-1} \det E_2^{-1} \dots \det E_p^{-1} \det B \\ &= \det E_1^{-1} \dots E_p^{-1} \det B \\ &= \det A \det B.\end{aligned}$$

**Case 2:**  $A$  is not invertible.

There must be a sequence of elementary matrices  $E_1, \dots, E_p$  such that  $A = E_1 \dots E_p U$  where  $U$  is the reduced row echelon form of  $A$ . Since  $A$  is not invertible, the invertible matrix theorem gives that  $U \neq I_n$ . Moreover,  $U$  must be upper triangular and have a zero along the diagonal. Thus  $\det A = \det E_1 \dots E_p U = \det E_1 \dots \det E_p \det U = 0$  since  $\det U = 0$ .

Since  $A$  is not invertible, we know  $AB$  is not invertible. Thus we can apply the same reasoning to  $AB$  to conclude  $\det AB = 0$ . Hence  $\det AB = 0 = \det A \det B$ .  $\square$

6. *Proof.* We prove the two implications.

$\implies$  Suppose  $A$  is invertible, then  $AA^{-1} = I_n$ . By the previous theorem,  $1 = \det A \det A^{-1}$ , and hence  $\det A \neq 0$ .

$\impliedby$  Suppose  $\det A \neq 0$ . We know there are elementary matrices  $E_1, \dots, E_p$  such that  $A = E_1 \dots E_p U$  where  $U$  is the reduced row echelon form of  $U$ . We know  $\det E_k \neq 0$  for all  $1 \leq k \leq p$ . Thus  $\det U \neq 0$ . Since  $U$  is an upper triangular matrix, we know the determinant of  $U$  is the product of the diagonal elements. Thus  $U$  cannot have a zero along the diagonal. Thus  $U$  has all ones along the diagonal. Since  $U$  is a square matrix, we get  $U = I_n$ . By the invertible matrix theorem,  $A$  must be invertible.  $\square$

7. (a) **True.** Since  $A$  is invertible, we know  $AA^{-1} = I_n$ . Thus  $1 = \det I_n = \det AA^{-1} = \det A \det A^{-1}$ , and hence  $\det A^{-1} = \frac{1}{\det A}$ .

(b) **True.** Let  $a_{ij}$  be the entry in row  $i$  and column  $j$  of  $A$  and let  $b_{ij}$  be the entry in row  $i$  and column  $j$  of  $A^T$ .

Expand the determinant of  $A$  along the first column and use  $a_{ij} = b_{ji}$ ; namely,

$$\begin{aligned}\det A &= a_{11} \det A_{11} - a_{21} \det A_{21} + \dots + (-1)^{n+1} a_{(n+1)1} \det A_{(n+1)1} \\ &= b_{11} \det(A^T)_{11} - b_{12} \det(A^T)_{12} + \dots + (-1)^{n+1} b_{1n} \det(A^T)_{1n} = \det A^T.\end{aligned}$$

(c) **False.** Let  $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$  and  $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ , then  $A \sim B$  and  $\det A = 1$  and  $\det B = -1$ .

8. Let  $M_{n \times n}$  be the set of  $n \times n$  matrices, then  $\det : M_{n \times n} \rightarrow \mathbb{R}$  is a function. This function fails to be linear since  $\det I_2 = 1$  but  $\det 2I_2 = 4$ .

It is worth noting that if we instead defined the function  $f_j : \mathbb{R}^n \rightarrow \mathbb{R}$  by  $\det [\vec{a}_1 \dots, a_{j-1} \vec{x} \vec{a}_{j+1} \dots \vec{a}_n]$ , then  $f_j$  is linear. Thus the determinant is linear in each column; we call this property **multi-linear**.